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Pitfalls of Disaster Information Sharing with Non-Japanese People

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Abstract

Many parts of Japan have experienced devastating earthquakes and tsunami in the past 30 years. How did emergency information reach non-Japanese residents and international tourists who did not understand Japanese language in the areas stricken by those earthquakes? In this paper the authors discuss how the way of disseminating disaster information in multiple languages has developed through the disaster experiences and what kind of pitfalls were revealed in each earthquake disaster.

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1. Introduction

The population of non-Japanese people has been increasing and Japan is at present home to over 3 million non-Japanese (2023)ⁱ. International tourists also continue to increase, with the arrivals of 25 million in 2023, having almost recovered to pre-pandemic levelsⁱⁱ. The Noto Peninsula earthquake that struck on New Year's Day of 2024 reminded us that Japan is a country with a high number of natural disasters, from earthquakes to typhoons. As Japan becomes a multicultural and diverse society the need for inclusive disaster preparedness and responses also grows.

Many parts of the country have experienced devastating earthquakes and tsunami in the past 30 years, such as Great Hanshin Earthquake in 1995, the Great East Japan Earthquake in 2011, Kyushu (Kumamoto) Earthquake in 2016 and Hokkaido Eastern Iburi Earthquake in 2018. How did emergency information reach non-Japanese people who did not

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understand Japanese language in the areas stricken by those earthquakes? In this paper the authors discuss how the way of disseminating disaster information for foreigners has developed through the earthquake disaster experiences and what kind of pitfalls were revealed. Also the authors analyze the pitfalls revealed in each case concerning disaster information provided with consideration for non-Japanese, and describe those pitfalls using system thinking approach.

Below, Section 2 describes related research. Next, Section 3 explains case studies to show pitfalls of disaster information sharing. Section 4 describes pitfall patterns in Systemigram. In Section 5, we discuss our considerations, and in Section 6, we present a summary and future issues.

2. Related work

2.1. Disaster information sharing

The Japan Tourism Agency is developing a push-type information dissemination app that provides disaster information using ICT (information and communication technology) and environmental improvements by local communities and private businesses responsible for accepting foreign tourists visiting Japan to ensure safety in the event of a disaster. It provides “Safety tips [1].” Hata [2] points out that, as an issue in providing disaster information to foreigners visiting Japan, “It is impossible to get foreigners visiting Japan to install disaster information providing apps developed by national and local governments.”

The Japan Tourism Agency's “Guidebook for creating tourism crisis management plans” [3] assumes four phases: “disaster mitigation,” “crisis preparation,” “crisis response,” and “crisis recovery.” It presents templates and worksheets for each type of “municipality/DMO (Disaster Management Office)” and “business operator (tourism, food, accommodation, transportation).” The templates are the main body of the guide, reference examples, and guidelines for disseminating information in times of crisis and disaster. The worksheet is a template for tourism crisis management plans, etc.

2.2. System thinking

Checkland proposed Soft System Methodology (SSM) [4] to analyze complex human activities. System model diagram is used in SSM. Since individual activities included in a human activity system can be expressed by “verbs,” in order to define a human activity system, it is necessary to be able to express the “connectivity” between verbs. In SSM, this type of relationship between activities is called a conceptual model. The components of a conceptual model consist of activities, logical dependencies between activities, external inputs and outputs, and outputs to control activities. System model diagram is a diagram that describes the conceptual model of SSM.

Boadman [5,6] focused on the fact that SSM system model diagrams easily correspond to natural language expressions, and proposed the Systemigram as a diagram that more clearly corresponds to natural language sentences. Systemigram allows you to express the structure of systems and problems in an easy-to-understand manner. In Systemigram, noun phrases are used as nodes, and relationships between nodes are defined by verb phrases that indicate the relationships between noun phrases. Systemigram is also used to analyze complex systems. For example, Systemigram is being used to analyze the complexity of enterprise architectures and SoS (System of Systems).

Mussante, Frittmann, and Edson [7] proposed a disaster management template that generalized the results of analyzing disaster management during past large earthquakes using Systemigram. They also report that the disaster template can be applied to disaster management during other earthquakes to identify important success factors for disaster management. The identified factors are Actors, Aid and Delivery, Area of Success, Event, Issues, Partial success, and Vulnerable sectors.

3. Case studies

The following studies clarifies pitfalls of disaster information sharing according to four Japanese earthquake cases. Table 1 shows the occurrence and human damage of the four cases.

Table 1. Japanese Earthquake cases.

Earthquake	Occurrence	Human damage
The Great Hanshin-Awaji, 1995	January 17, 1995(magnitude 7.3)	6,434 dead, 3 missing (As of May 2006)
The Great East Japan, 2011	March 11 (over magnitude 7.0: 6times; over 6.0: 96 times)	22,212 dead and missing (As of March, 2023)
Kumamoto, 2018	April 14 (magnitude 6.5) April 16 (magnitude 7.3)	211 dead, 1142 severely injured, 1604 slightly injured (As of March, 2017)
Hokkaido Eastern Iburi, 2018	September 6, (magnitude 6.7)	44 dead, 785 injured (As of March 31, 2019)

3.1. The Great Hanshin-Awaji Earthquake

The Great Hanshin-Awaji Earthquake occurred on January 17 in 1995, in the southern part of Hyogo Prefecture, in Japan's main island of Honshu. More than 6,400 people were killed including approximately 200 non-Japanese people in and around Kobe City. On that occasion, non-Japanese people who did not understand the Japanese disaster terminology found it difficult to get relevant information. The serious impact of the language barrier was after the quake occurred. They did not know about evacuation sites, information about the aftershocks, when and where to get food or water, and so on.

The Great Hanshin-Awaji Earthquake served as a catalyst, making society aware of the presence of non-Japanese residents who couldn't understand Japanese. It became evident that multilingual information was necessary during emergencies. After the Earthquake, there has been a progress in the measures to support them in times of disaster. International affairs divisions of prefectures or big cities which hold a quite number of non-Japanese started to get prepared for the situation in order to provide necessary information in multiple languages.

Pitfalls: Due to a lack of awareness regarding the presence of foreign residents, there was little provision of information in multiple languages through mediums such as television, radio, and newspapers. Local government officials in disaster-affected areas, while recognizing the necessity of providing information in multiple languages, experienced delays in disseminating information to non-Japanese residents due to reasons such as time-consuming translation processes.

3.2. The Great East Japan Earthquake

In 2011, the Great East Japan Earthquake took place in the Tohoku Region, north-eastern part of Japan. The earthquake and particularly the ensuing tsunami killed nearly 20,000 people and caused a nuclear accident at a power plant in Fukushima Prefecture. Non-Japanese residents in the disaster areas were mainly students at Japanese language schools or universities, trainees and technical intern trainees and foreign wives married to Japanese men. Supporting measures for non-Japanese residents, such as the dissemination of disaster information in multiple languages, have been utilized at the time of this earthquake.

However, there has emerged a new issue. The support measures for non-Japanese people prepared after the 1995 Earthquake have been presumed to be utilized for foreigner-concentrated areas. Those measures have turned out to be less effective than expected in the case of the 2011 Earthquake due to sparsely scattered non-Japanese people in the disaster affected Tohoku Region.

Foreign wives live in local communities with their families isolated from the other non-Japanese residents and were placed in a hard situation. Most of them were unable to access internet information but only Japanese TV news and the disaster-related terms presented in Japanese were not understandable. Especially after the explosion accident at the Fukushima nuclear power plant, the information provided was difficult to comprehend. Foreign wives who tried to go back to their home countries leaving their Japanese husbands or children in Japan due to the recommendation for evacuation by their embassies or by their families in their home towns were given icy stares by people in their neighborhood or even by their Japanese families (Komai and Suzuki, 2012). What was most required among them was the mental care. But more miserable were non-Japanese wives who lost their Japanese husbands in tsunami. One

of the ways to help foreign wives not getting isolated is the formation of a network for those who are geographically scattered and also a network to get connected with Japanese residents.

Pitfalls: Disaster multilingual information for areas with concentrations of non-Japanese residents proved less useful in regions where non-Japanese residents were scattered. Most suffered non-Japanese residents were those who were married with Japanese husbands in the rural areas and could not access the multiple language information on the internet. What they needed most was not only disaster information but mental support.

3.3. Kumamoto Earthquake

The 2016 Kumamoto Earthquake occurred on April 14, 2016, with following several major earthquakes in the middle area of Kyushu Island, southwest of Japan. Non-Japanese residents in Kumamoto Prefecture, including permanent residents, spouses of Japanese, and others, evacuated to locations within Kumamoto Prefecture, such as to parks near their residences, to parking lots, to universities, or to other facilities that are used on a daily basis.

Kumamoto City International Center opened its facility as an emergency shelter on April 15, expecting to support non-Japanese residents in the area. However, international tourists from Korea, China, France, America and so on who were temporarily staying in Kumamoto also rushed to the center seeking for the multiple language information since they sought refuge in places where they could communicate with others. The employees of the Center were busy taking care of them, responding the inquiries and requests from them, and they managed to start the operation of Multilingual Support Center on April 23, a week after the earthquake occurred.

According to a survey conducted by the Survey Research Center (SRC) on the behavior of international tourists during the Kumamoto earthquake who were staying in the Kyushu region at the time of the earthquake, the top difficulty faced during the earthquake was the absence of an earthquake evacuation manual for foreigners, making it difficult to understand what actions to take (36.5%). This was followed by uncertainty about future travel arrangements (33.9%), disruption of all planned activities resulting in significant inconvenience (29.6%). Additionally, there was a lack of understanding about transportation information (19.1%).ⁱⁱⁱ

Pitfalls: Multilingual disaster information was designed with resident foreigners in mind. International tourists who were in the midst of their travels often had never experienced an earthquake, leaving them unsure of how to react. Furthermore, language barriers exacerbated the situation. Obtaining information about the status of their itinerary and transportation options for returning home proved challenging.

3.4. Hokkaido Eastern Iburi Earthquake

The Hokkaido Eastern Iburi Earthquake occurred on September 6, 2018, in the Hokkaido corner region, and its epicenter was located approximately 70 to 100 kilometers southeast from the capital city of Sapporo. The earthquake knocked out power of the entire region, affecting up to 2.95 million households. The tremor caused several power plants to shut down, disrupting the electrical frequency. This led to a chain reaction of other power plants to shut down. All public transport, including rail, subway, and bus services was shut down in Hokkaido due to the loss of electricity and damages. The quake disrupted television viewing in the area as well as telephone and internet connections.

According to the Sapporo City Tourism and MICE Promotion Department, it is estimated that at the time of the earthquake, there were over 23,000 tourists in the city, including approximately 5,000 international tourists.

Sapporo International Plaza (Convention Bureau) was designated to serve as a disaster multilingual support center during emergencies, providing information to foreigners. While measures were reasonably prepared, the situation proved to be a series of unforeseen events, including blackouts. A survey conducted by Survey Research Center Co., Ltd.

A survey conducted by Survey Research Center Co., Ltd. revealed that visitors affected by the Hokkaido earthquake expressed difficulties due to power outages causing a lack of information. However, 70% expressed a desire to revisit Japan. According to a survey which targeted 185 international tourists who were staying in Hokkaido at the time of the earthquake 185 individuals with proportions from countries such as China (30.8%), Taiwan (18.9%), Hong Kong (10.8%), and the United States (8.5%), about 40% expressed feeling endangered and being concerned about future earthquakes. Approximately 30% also expressed worries about their travel schedules. Many respondents cited power outages as a major issue during the earthquake. 67.0% reported not being able to obtain information due to power

outages and being unable to charge their smartphones. Nearly 30% encountered language barriers and were unsure where to go. Furthermore, about 70% stated that there was no prior explanation about evacuation guidance, with nearly half feeling that prompt action could have been taken if such information had been provided in advance.

Pitfalls: The dissemination of information on the internet proved ineffective during unforeseen power outages. Analog methods such as posters at stations were helpful for providing transportation information. International tourists who were relying on smartphones for information gathering found themselves cut off from updates. During emergencies, it is essential to provide information in multiple languages through various media channels to reach a wide audience.

4. Pitfall analysis using Systemigram

The Fig. 1 through Fig.4 show the pitfalls regarding information sharing with non-Japanese people during the four major earthquakes using the Systemigram. These diagrams were written based on Boardman's method of describing Systemigrams. The nodes of a Systemigram are nouns representing subjects and objects, actors, abstract processes, and knowledge. Node relationships are derived from verb phrases that express the relationships between nouns of subjects and objects.

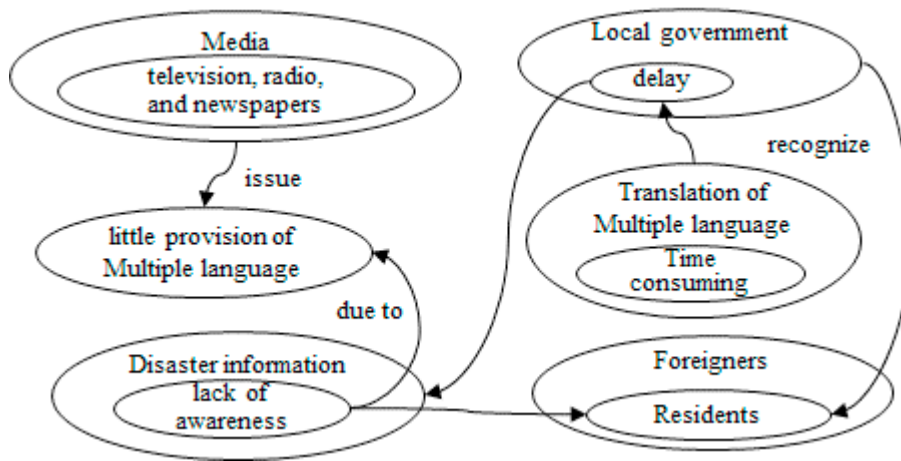


Fig. 1. Systemigram for Great Hanshi- Awaji.

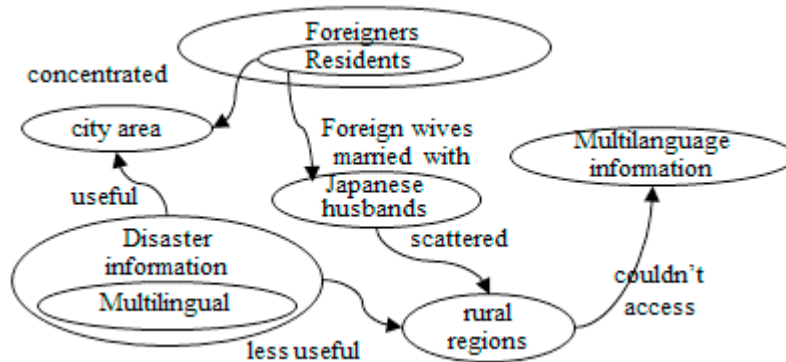


Fig. 2. Systemigram for Great Eastern Japan.

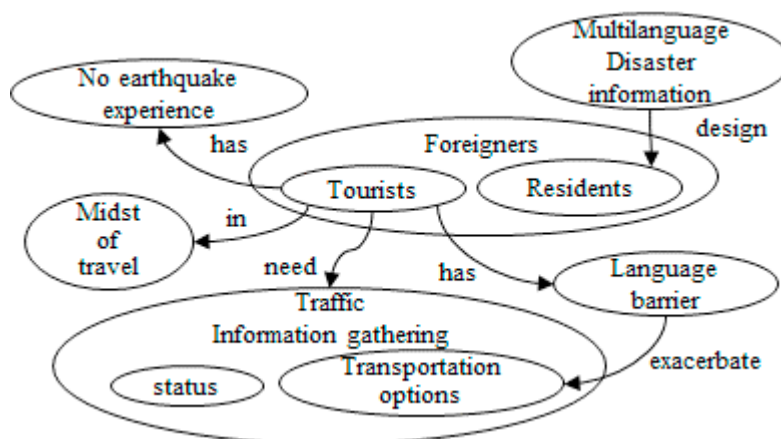


Fig. 3. Systemigram for Kumamoto.

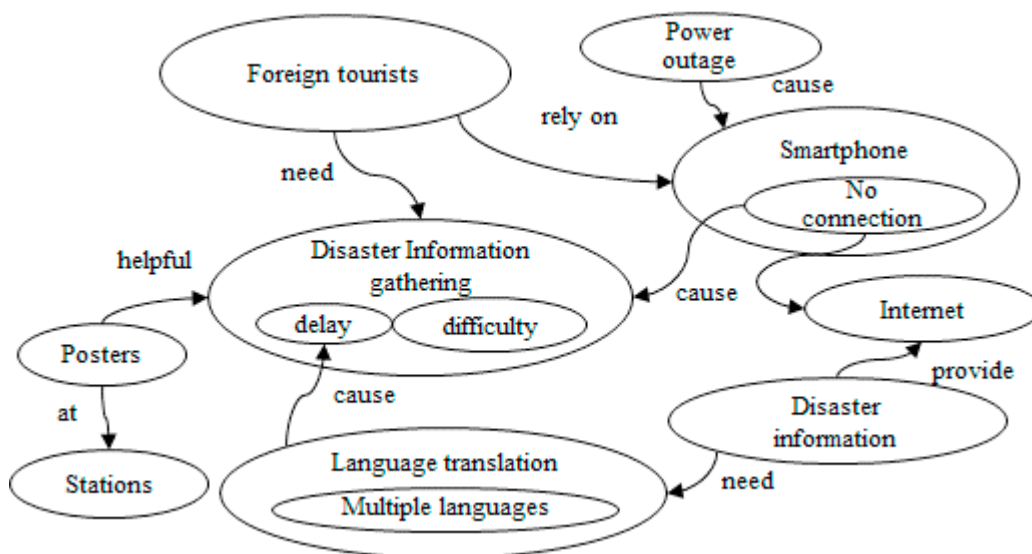


Fig. 4. Systemigram for Hokkaido Eastern-Iburi.

The four figures above facilitate an understanding of the types of pitfalls that became evident during each earthquake occurrence. Fig. 5 integrates pitfalls in information provision revealed during the four earthquake occurrences.

Most elements of pitfalls revealed in the four earthquakes remain in the integrated pitfalls. This indicates that strategies for assisting foreign victims differ depending on the target audience for information provision during earthquake disasters. Furthermore, the Systemigram provides an efficient visual medium for integrating various pitfalls.

This Fig. 5 Systemigram can be considered an efficient visual medium, as it combines and integrates different pitfalls from four past earthquakes into a single diagram. Furthermore, the fact that each element of the pitfalls is different indicates the impossibility of generalization. There are six categories (Actor, Activity, Information, Means, Situation, Location) in the nodes. Table 2 presents examples of category nodes identified in the four earthquake cases.

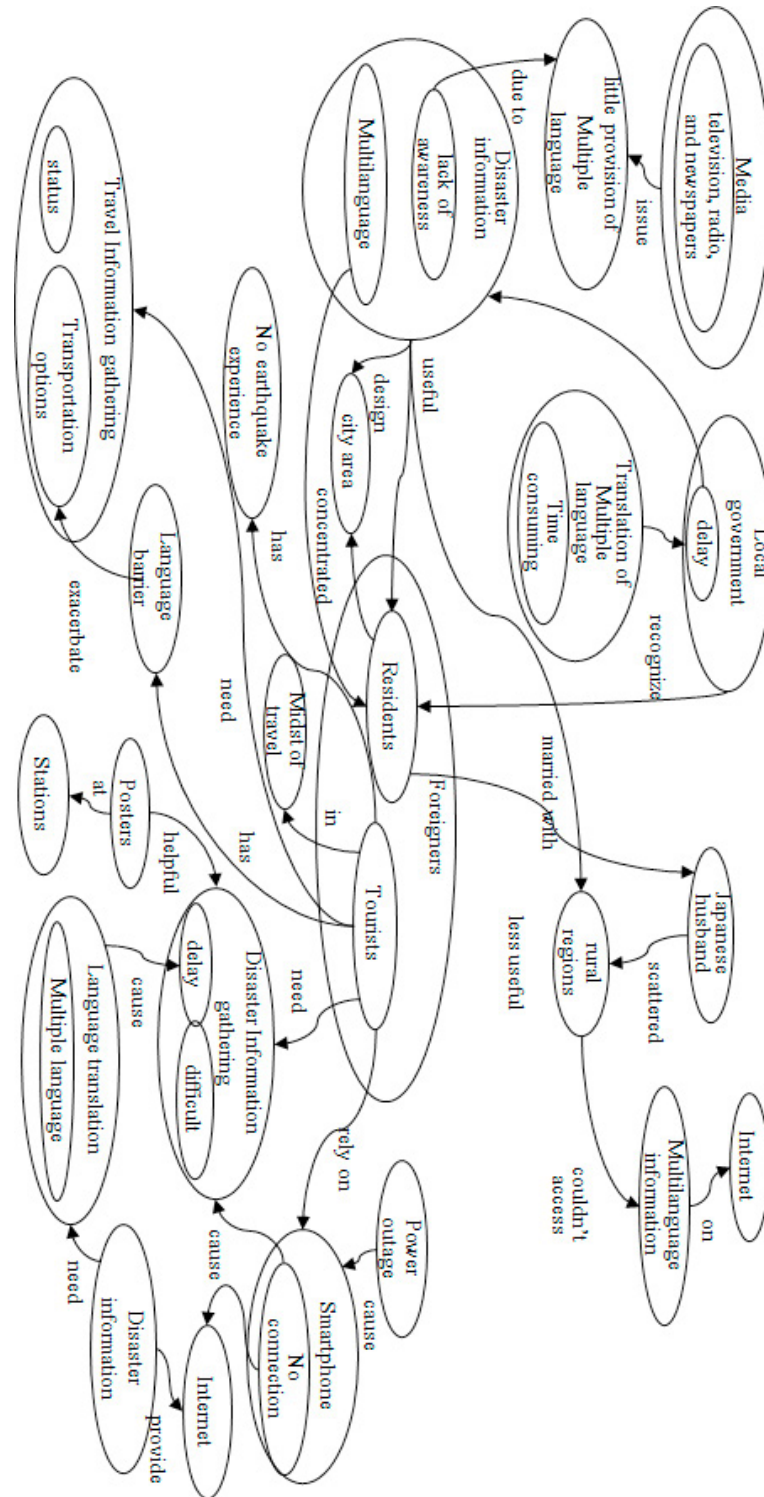


Fig. 5. Systemigram for Integrated Pitfalls.

According to the above applications of Systemigram, there are six categories of nodes, i.e., Actor, Activity, Information, Means, Situation, and Location. Table 2 shows examples identified category nodes in four earthquake cases.

Table 2. Node Category examples of Japanese Earthquake cases.

Earthquake	The Great Hanshin-Awaji	The Great East Japan	Kumamoto	Hokkaido Eastern Iburi
Actor	Local government, Foreigners, Residents	Foreigners, Residents Japanese husbands	Foreigners, Residents Tourists	Foreign tourists
Activity	Translation into multiple languages	--	No earthquake experiences Traffic information gathering	Language translation
Information	Disaster information	Multilanguage Disaster information	Multilanguage Disaster information Transportation options	Disaster information Multiple language
Means	Media	Internet	--	Internet, Smartphone Posters
Situation	Little provision of Multiple Language Information Lack of awareness about the existence of non-Japanese	--	Midst of travel Language barrier, status	Power outage No connection Delay, difficult
Location	--	City area, Rural region	--	stations

5. Discussions

This section discusses the novelty, effectiveness, and limitations of the proposed approach.

5.1. Novelty

So far, there hasn't been an analysis of pitfalls in disaster information sharing for vulnerable populations using Systemigrams. There hasn't been a common method for visually representing pitfalls in disaster information sharing comprehensively. We identified six elements—Actor, Activity, Information, Means, Situation, and Location—to describe these pitfalls, as shown in Table 2. By using Table 2, it's possible to analyze the pitfalls in disaster information sharing.

5.2. Effectiveness

The examples in section 4 shows the applicability of Systemigram to analyze pitfalls in sharing disaster information. Furthermore, the four Systemigrams derived from earthquake cases in Japan have been seamlessly integrated into a single comprehensive diagram. This demonstrates flexibility in considering better information provision while indicating each respective pitfall. It's also noteworthy that the six constituent elements in Table 2 serve as common factors for describing Systemigrams. Pitfalls in disaster information sharing arise from these common constituent elements.

5.3. Limitations

Since our analysis in this paper focused on four earthquake incidents, it is evident that investigating other cases of earthquakes is necessary to develop a comprehensive model of pitfalls in disaster information for foreign audiences. Furthermore, as the integrated pitfalls diagram in Figure 5 only utilizes specific instances of pitfalls, there is a need to clarify general patterns of pitfalls in disaster information sharing. Additionally, we need to examine pitfalls in information sharing during disasters beyond the cases of the four earthquakes discussed. Then, it is necessary to

generalize patterns of pitfalls that occur during disaster information sharing. Also, there is a need for quantitative evaluation not only of earthquake-related pitfalls but also of other natural disaster pitfalls.

The pitfalls in disaster information sharing indicate a lack of crucial information constituent elements. It would be effective for disaster information sharing platforms to include measures to prevent such pitfalls. However, this paper did not mention disaster information sharing data models as a countermeasure to these pitfalls. It is necessary to clarify these data models to mitigate the risks of information sharing.

The relationship between the disaster information sharing diagram and the six category nodes also needs to be examined. While constituent elements have been identified, the mapping from constituent element nodes to the diagram is not yet clear.

6. Summary

At the Great Hanshin-Awaji Earthquake in 1995, foreign residents became information vulnerable, leading to the recognition of the need for multilingual information provision for those who couldn't understand Japanese. This marked the beginning of acknowledging the necessity for "building a multicultural coexistence society" as a societal challenge. Subsequently, mechanisms for multilingual information provision, especially during emergencies, began to be prepared by regional international exchange associations.

During the 2011 Great East Japan Earthquake, international exchange associations in affected areas provided information and support to foreign residents. However, unlike the Hanshin region, where foreign residents were concentrated, it was difficult to grasp their existence in the East Japan region, causing information dissemination challenges. During the 2016 Kumamoto Earthquake, international tourists present were confused about what actions to take, facing language barriers and flooding international exchange centers where communication in foreign languages was possible. Moreover, the information sought by tourists mainly pertained to transportation, differing from that needed by foreign residents.

During the 2018 Hokkaido Eastern Iburi Earthquake, power outages left international tourists without local knowledge relying on the internet, resulting in a complete information blackout. While information dissemination is crucial, there's an urgent need for mechanisms ensuring that pertinent information reaches those who need it accurately and promptly.

Simply categorizing someone as a foreigner doesn't adequately encompass the diverse range of their backgrounds and experiences with disasters prior to coming to Japan.

The concept proposed by American sociologist Moore to explain disaster response within communities is known as "disaster culture." Awareness and preparedness for specific disasters such as earthquakes, floods, and typhoons vary depending on the region or country. Disaster culture can also be shaped through media coverage and education. Adequate information dissemination and education contribute to enhancing understanding and response capabilities to disasters. While disaster culture exhibits different characteristics depending on the region or country, its presence is crucial for bolstering overall societal resilience against disasters.

This paper examines the evolution of information-sharing behaviors during disasters in Japan, using earthquake disasters as a case study, from the perspective of systems theory. The pitfalls that have been overlooked during disaster information sharing have not been analyzed using Systemigram before. By comprehensively visualizing the pitfalls in disaster information sharing in this paper, it becomes possible to consider strategies and means of information dissemination to information disadvantaged individuals as lessons learned. Merely, disseminating information in multiple languages may not suffice. However, having information available in languages people understand, and having individuals who speak those languages, can provide a significant sense of reassurance.

Acknowledgements

Kumi Sato [8, 9] conducted interviews with individuals responsible for providing assistance to non-Japanese residents and tourists, such as the International Exchange Association, in the four earthquake-affected regions outlined in this paper. We sincerely appreciate their cooperation and assistance.

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ⁱ The number of foreign nationals residing in Japan hit a record high of over 3.2 million in 2023, government data has shown, as more workers entered to meet demand created by the country's labor shortage.

ⁱⁱ Japan received 25 million tourists in 2023, reaching 79% of pre-COVID level, according to official estimates released by the Japan National Tourism Organization.

ⁱⁱⁱ Source: SURVEY RESEARCH CENTER CO., LTD. From April 21st to 22nd, 2016, interviews were conducted with 115 foreign tourists at the departure lobby of the Fukuoka Airport International Terminal. https://www.surece.co.jp/wp_surece/wp-content/uploads/2017/10/kumamoto_overview.pdf